



J. Dairy Sci. TBC

<https://doi.org/10.3168/jds.2024-26216>

© TBC, The Authors. Published by Elsevier Inc. on behalf of the American Dairy Science Association®.
This is an open access article under the CC BY license (<https://creativecommons.org/licenses/by/4.0/>).

Predicting postpartum diseases in Holstein cows using milk spectra and machine learning—Retrospective assessment from diagnosis date

D. Lin,¹ J. Li,^{2,3,4*} J. A. Seminara,¹ and J. A. A. McArt^{1*†}

¹Department of Population Medicine and Diagnostic Sciences, College of Veterinary Medicine, Cornell University, Ithaca, NY 14853

²City University of Hong Kong, Shenzhen Research Institute, Shenzhen, China

³Department of Infectious Diseases and Public Health, Jockey Club College of Veterinary Medicine and Life Sciences, City University of Hong Kong, Hong Kong, China

⁴School of Data Science, City University of Hong Kong, Hong Kong, China

ABSTRACT

Dairy cows commonly experience health disorders in the early-lactation period. Although Fourier-transform infrared (FTIR) spectroscopy offers a noninvasive and cost-effective method for analyzing milk composition, its potential in predicting subsequent early-lactation diseases has yet to be adequately explored. This study aimed to uncover the ability of milk FTIR spectra to predict postpartum diseases in 1,162 Holstein cows from a commercial dairy farm in Cayuga County, NY. We collected proportional milk samples daily on cows in the early-lactation pen and stored milk at 4°C until analysis via FTIR spectroscopy. Cows were monitored through 30 DIM and classified as healthy ($n = 825$; no adverse health events) or diseased ($n = 311$; diagnosis of clinical ketosis, metritis, displaced abomasum, or mastitis, or any combination of these). We developed predictive models for 8 distinct time periods preceding the diagnosis date (>10 d, 10–8 d, 7–6 d, 5–4 d, 3 d, 2 d, 1 d, and 0 d), using regression, machine learning, and deep-learning methods applied to milk FTIR spectral data. Model performance was evaluated through a repeated down-sampled double cross-validation framework and permutation tests. Our results showed that progressive changes in spectral regions related to the absorbance peaks of fat, protein, and lactose are correlated with disease progression, leading to an increase in average area under the receiver operating characteristic curve (AUROC) from 0.50 (>10 d before diagnosis) to 0.72 (1 d prior) and 0.76 (the day of diagnosis) across all model types. Partial least squares-discriminant analysis (PLS-DA) models using milk FTIR spectra achieved an average AUROC of 0.71 from 7 d

before diagnosis, outperforming models based on cow-level features (0.62) or combined with spectra-predicted milk major components (0.67). Among spectral models, PLS-DA reached the highest average AUROC (0.74), followed by long short-term memory (0.72), and surpassed ridge regression (0.71) and random forest (0.69). These findings highlight the effectiveness of using milk FTIR spectra to predict upcoming health conditions in early-lactation Holstein dairy cows, although broader evaluation is necessary to assess generalizability and on-farm utility.

Key words: milk spectra, Fourier-transform infrared spectroscopy, machine learning, deep learning

INTRODUCTION

Early lactation is a critical period for the health, fertility, and productivity of dairy cows. During this time, energy deficit can lead to excessive fat mobilization, increasing the risk of metabolic and infectious diseases such as hyperketonemia, metritis, retained placenta, displaced abomasum (**DAb**), and mastitis (Hammon et al., 2006; Chapinal et al., 2011; Hertl et al., 2018). Currently, blood metabolic profiling serves as the most common method to monitor the health status of dairy cows in early-lactation period (Overton et al., 2017). Several blood markers, including glucose, nonesterified fatty acids, and BHB are widely used as key indicators of diseases (Leblanc 2010; McArt et al., 2012; Esposito et al., 2014). Although effective, blood sampling and the associated analysis are costly, labor-intensive, and invasive, limiting their large-scale and daily application. This highlights the need to develop and apply new methods to monitor the health status of early-lactation dairy cows.

There is increasing evidence suggesting variations in milk composition mirror the change in health status of dairy cows (Hamann and Krömker 1997; Overton

Received December 23, 2024.

Accepted September 8, 2025.

*These authors contributed equally to this work.

†Corresponding authors: jun.li@cityu.edu.hk and jmcart@cornell.edu

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-25. Nonstandard abbreviations are available in the Notes.

et al., 2017; Gross and Bruckmaier 2019), and milk analysis has shown evident advantages in practice due to its noninvasiveness and ability to be incorporated into current standard milking procedures (Giannuzzi et al., 2023). Fourier-transform infrared (FTIR) spectroscopic analysis of milk is a rapid and cost-effective method that has been routinely used for quantification of milk compositions (i.e., fat, protein, CN, lactose, and urea; Luinge et al., 1993; De Marchi et al., 2014; Grelet et al., 2015), and in recent years, FTIR spectroscopy of milk has gained attention as a noninvasive tool for predicting early-lactation health outcomes in dairy cows. Various studies have indicated its effectiveness in predicting energy status (McParland et al., 2011; McParland et al., 2014), with FTIR-determined milk fatty acids and BHB linked to energy deficit and hyperketonemia, yielding area under the receiver operating characteristic curve (AUROC) of 77% to 82% (Churakov et al., 2021) and sensitivities of 76% to 82% (Van Kneegsel et al., 2010; van der Drift et al., 2012; Bonfatti et al., 2019). This technology has also been applied to mastitis detection, achieving sensitivities of 50% to 95% and specificities of 32% to 85% (Soyeurt et al., 2012; Dale et al., 2019; Rienesl et al., 2022), as well as differentiation of normocalcemic cows from those with subclinical hypocalcemia (Seminara et al., 2023). Despite these promising results, some key limitations still persist. Most existing studies have relied on infrequent milk sampling (e.g., monthly or sporadic), which limits the ability to capture spectral dynamics preceding disease onset, thereby limiting FTIR spectroscopy's potential in predicting upcoming disease events. Although Bach et al. (2019) collected milk at 4 time points between 3 and 18 DIM and found FTIR spectroscopic-predicted milk components were promising indicators of subsequent disease or removal, predictive models integrating raw milk spectra with cow-level information are still in their infancy, and the crucial spectral regions contributing to the prediction also remain unidentified.

Our goal was to advance our understanding of milk FTIR spectroscopic analysis and determine its potential utility in predicting upcoming disease in early-lactation dairy cows using a series of modeling techniques, where “disease” represents a combined group of multiple postpartum conditions with potentially heterogeneous spectral patterns. Our specific objectives were to (1) evaluate whether milk spectra become increasingly abnormal and more beneficial for prediction as disease diagnosis nears, (2) explore the performance of various machine learning and deep-learning methods using milk FTIR spectral data, milk yield, SCC, and parity in predicting disease within the first 30 DIM, and (3) determine which spectral regions are most informative for prediction.

MATERIALS AND METHODS

Study Approval, Design, and Population

Our research project was reviewed by the Cornell University Institutional Care and Use Committee and determined not to meet the regulatory Cornell Policy 1.4 “Description for use of animals for research, teaching, or testing” because there was no animal contact or manipulation. Subsequently, our project research application underwent ethical and scientific review and was approved by the Cornell University Veterinary Clinical Studies Committee (CUVCSC Protocol ID#062420–05).

We conducted a prospective observational cohort study from June through August 2021 on a single commercial dairy farm in Cayuga County, NY. An informed consent letter approving participation in our study was signed by the farm owner. This farm was selected based on proximity to the Cornell Ambulatory and Production Medicine Clinic (Ithaca, NY) and its history of successful prior participation in research studies. Our project was designed, analyzed, and written following the STROBEvet guidelines for strengthening the reporting of observational studies in veterinary epidemiology (Sargeant et al., 2016).

The farm housed approximately 4,400 lactating cows which were milked 3 times a day via a 100-stall rotary parlor (DeLaval). During the study period, cows averaged 43.5 kg of milk per cow per day. Lactating cows were housed in 10 freestall pens and bedded with recycled manure solids. The early-lactation pen, a 3-row design, commingled primiparous and multiparous cows which were moved from the maternity pen after their first milking within 8 h of calving. Cows remained in the early-lactation pen until moved to a high production pen between 7 and 21 DIM. Timing of movement out of the early-lactation pen was based on management decisions regarding parity, daily milk yield, and pen stocking density. The early-lactation pen was fed once daily at 0745 h and milked at 0230, 1030, and 1830 h. Ingredients and formulated nutrient composition of the early-lactation TMR can be found in Callero et al. (2023).

Animal Sampling and Farm Data Extraction

Cows in the early-lactation pen at any point during our study period were eligible for inclusion. On Monday through Friday, we collected proportional milk samples (DeLaval Fat Sampler MM25–27 BC; DeLaval) from all cows in the early-lactation pen once daily at the 1030 h milking. After automatic retraction of the milking unit at the conclusion of each cow's milking, the proportional sampler cup was removed, and milk was inverted 3 times

before being poured into 60-mL plastic vials without preservatives (Aptar CSP Technologies). At the conclusion of the 1030 h early-lactation pen milking shift (approximately 1100 h), we placed the milk samples in ice-filled coolers and transported samples at 4°C to the Barbano Laboratory in the Department of Food Science at Cornell University (Ithaca, NY) for FTIR spectroscopic analysis and SCC determination, which was measured and calibrated as described previously (Bach et al., 2021).

Sixty days after the conclusion of the sample-collection portion of the study, we extracted disease data from the farm's software recording system (DairyComp 305; VAS). Diseases of interest included farm-diagnosed ketosis, metritis, DAB, and mastitis within 30 DIM and were defined based on the following farm protocols: ketosis, a blood BHB concentration ≥ 1.2 mmol/L using a handheld device (Precision Xtra; Abbott) on cows flagged by computer software for physical examination due to a decrease in milk yield; metritis, a cow with presence of foul-smelling brown to red vaginal discharge and a rectal temperature $\geq 39.7^\circ\text{C}$; DA, auscultation of a ping sound on simultaneous percussion over the left or right abdomen in a line extending from the tuber coxae to the olecranon on cows flagged by computer software for physical examination due to a decrease in milk yield; and mastitis, based on visual identification of a hot and swollen quarter and presence of abnormal milk with or without garget. A binary variable was created for the target health outcome for each cow, and cows diagnosed with at least one disease within the first 30 DIM were labeled "diseased," whereas cows without these diagnoses in the same period were considered "healthy."

Initially, we obtained 7,142 milk samples with valid FTIR spectra from 1,162 early-lactation cows. We excluded 88 samples ($n = 51$ cows) collected after 30 DIM, and 728 samples ($n = 125$ cows) collected after the diagnosis of at least one disease (ketosis, metritis, DA, or mastitis). This resulted in a final data set of 6,326 milk samples ($n = 1,136$ cows): 1,306 samples ($n = 311$ cows) in the disease group and 5,020 samples ($n = 825$ cows) in the healthy group. An illustration of cows and sample enrollment is shown in Figure 1A.

Milk Spectra Preprocessing

Milk spectra were measured with a FTIR spectrophotometer (Lactoscope model CombiScope FT600, Delta Instruments, Drachten, the Netherlands) within 0.6 ± 0.7 d of collection as described by Callero et al. (2023). Each spectrum consisted of 935 absorbance values ($\log[1/\text{transmittance}]$) between 4,000 and 990 cm^{-1} with a reso-

lution of 8 cm^{-1} . For the spectra, highly noisy regions ($<3,000$ cm^{-1} ; $>1,000$ cm^{-1} ; 1,700–1,585 cm^{-1}), which is the consequence of a high CaF_2 and water absorption, were first removed (Heuer et al., 2001). To improve the signal-to-noise ratio, multiple measurements of spectra were then averaged for each sample (Subramanian et al., 2009; Pazzola et al., 2023). Most milk samples had 2 replicate measurements, with the distribution of replicates as follows: 1 (4 samples), 2 (6,264 samples), 3 (26 samples), 4 (31 samples), and 5 (1 sample). Finally, each individual milk FTIR spectrum was min-max normalized to a [0,1] range to boost wavenumbers with potentially important information for modeling. The illustration of milk spectra preprocessing procedures is shown in Figure 1A.

Based on the absorption of different milk components, the spectra were split into 4 regions: R1, spanning 3,000–2,800 cm^{-1} , mainly consisting of symmetrical and asymmetrical CH_2 -stretch peaks at 2,923 cm^{-1} and 2,854 cm^{-1} respectively, related to fat absorption; R2, covering 1,800–1,700 cm^{-1} , also related to fat absorption; R3, spanning 1,585–1,000 cm^{-1} , serving as the fingerprint area of protein (1,540 cm^{-1}), lactose (1,040 cm^{-1}) and fat (1,461 cm^{-1} and 1,157 cm^{-1}); and R4, the wavenumber regions covering 2,800–1,800 cm^{-1} , typically regarded as nonspecific signals.

Modeling

Framework. We adopted a repeated down-sampled double cross-validation framework in this study. First, to avoid biased results from the overrepresented healthy group (Tyagi and Mittal 2020; Susan and Kumar 2021), we randomly down-sampled cow-level samples from the healthy group an equal number of samples to that of the disease group. To account for potential confounding by lactation stage, each down-sampled healthy subset was matched to the disease group by DIM distribution, ensuring both groups had comparable DIM profiles (Supplemental Figure S1, see Notes). This random sub-sampling process was repeated 50 times to ensure thorough utilization of data from the healthy group. Then, for each balanced dataset, we applied a repeated double cross-validation strategy to integrate both model selection and performance evaluation (Filzmoser et al., 2009; Szymańska et al., 2012). This procedure involves 2 nested cross-validation loops: an inner loop for model selection and an outer loop for model assessment. In the outer loop, the dataset was randomly partitioned into training and test sets, with the test set held out exclusively for final evaluation. To prevent data leakage, cross-validation was performed at the cow level (based on cow ID), en-

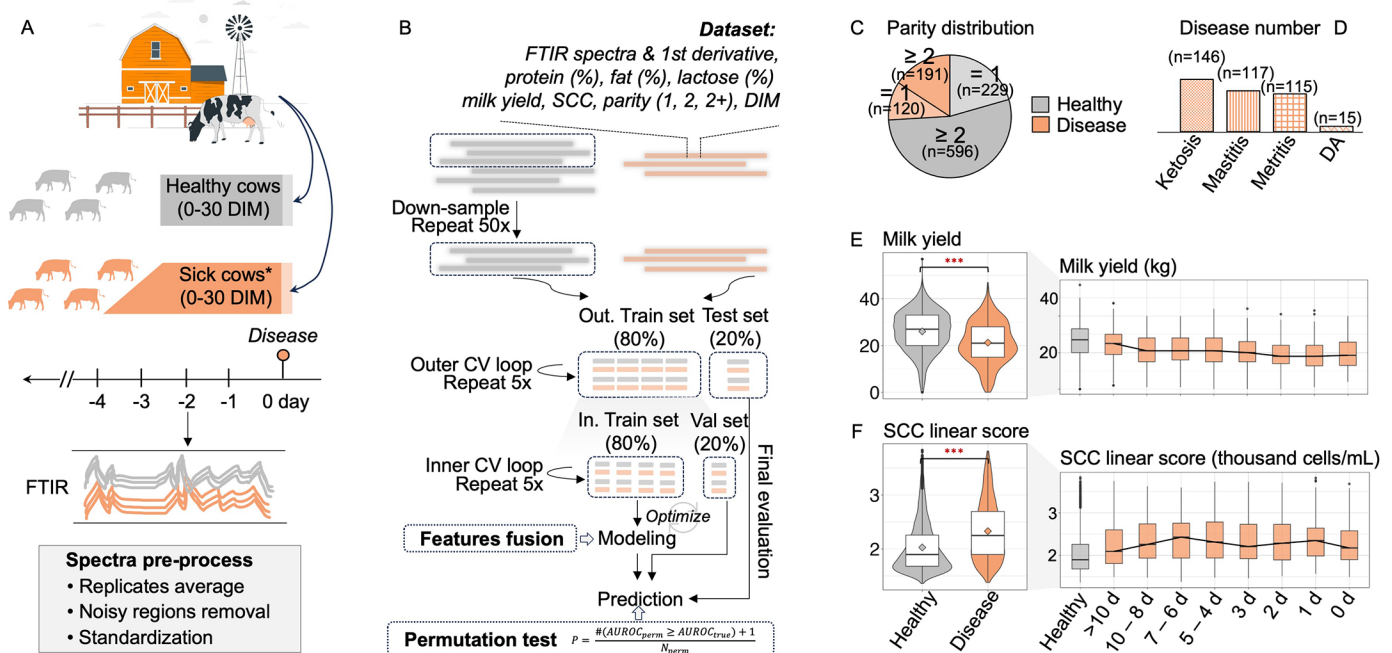


Figure 1. Illustration of the study design and dataset. (A) Retrospective evaluation of milk FTIR spectroscopy for disease prediction within 30 DIM. This study included 5,020 milk samples in the healthy group ($n = 825$; no adverse health events) and 1,306 milk samples in the disease group ($n = 311$; diagnosis of clinical ketosis, metritis, DAB, or mastitis, or any combination thereof) from a single commercial farm in central New York. (B) Schematic of the repeated down-sampled double cross-validation and feature fusion strategy for integrating spectral and cow-level static features. Out. = outer, Train = training, In. = inner, Val = validation. (C) Parity distribution in healthy and disease groups. (D) Case incidence of each disease. (E, F) Milk yield (kg) per milking (E) and linear score SCC: $\log_{10}(\text{SCC})$ in thousand cells/mL (F) distributions for healthy and disease groups across various time points before disease diagnosis. * $P < 0.05$. ** $P < 0.01$. *** $P < 0.001$.

ensuring that samples from the same individual were not split across training and test sets within any fold. Within the inner loop, the training set was further split into inner training and validation subsets. The inner training set was used to fit models under varying hyperparameter configurations (e.g., number of components, trees, or neural network layers), and the validation set was used to identify the optimal configuration. The model with the best inner-loop performance was then evaluated on the test set from the outer loop. This outer-loop procedure was repeated until every cow-level sample had served as a test instance exactly once, ensuring that the performance estimates reflect generalization to unseen data. In our study, both the inner and outer cross-validation loops were implemented as 5-fold cross-validation, with a 4:1 split of training to validation or test within each loop, to balance estimation reliability and runtime efficiency under a computationally intensive repeated double cross-validation scheme. The entire double cross-validation procedure was then repeated 5 times with different random partitions to enhance robustness and mitigate variability due to data partitioning. The repeated down-sampled double cross-validation framework is illustrated in Figure 1B.

Models. Predictive models were constructed using Scikit-learn (Pedregosa et al., 2012). To find the model with the best performance, we evaluated a branch of classifiers, including regression-based models (partial least squares-discriminant analysis [PLS-DA] and ridge regression), tree-based model (random forest [RF]) and deep-learning model (long short-term memory [LSTM]).

PLS-DA. As a classic method for spectra analysis, PLS-DA (Barker and Rayens, 2003) was set up as the baseline to compare the predictive performance between different models. It was fitted using the `PLSRRegression()` in Scikit-learn. If a cow's predictive probability was greater than 0.5, the cow was classified as 1 (disease group); otherwise, the cow was considered as 0 (healthy group). To determine the optimal number of latent variables, a hyperparameter search was conducted by varying the number of components (`n_components`) from 2 to 20 in steps of 2.

Ridge Regression

Ridge regression (Hoerl and Kennard, 1970) is a penalized regression model that applies L2 regularization to address multicollinearity in high-dimensional spectral

data. It was implemented using the RidgeClassifier() function in Scikit-learn. The optimal value of α was determined by searching over the range 0.5 to 1.0 with a step size of 0.1.

RF

Random forest is an ensemble method based on bagged decision trees, capable of modeling nonlinear relationships and handling the complexity of high-dimensional spectral data (Breiman 2001). It was built using the RandomForestClassifier() function in Scikit-learn. To optimize model performance, a random search was conducted over the number of trees ($n_estimators$), evaluated at 10, 20, 30, 40 and 50.

LSTM

Long short-term memory, a type of recurrent neural network (RNN), uses memory gates to capture long-term dependencies and address vanishing gradient issues (Hochreiter and Schmidhuber 1997). Its ability to model sequential data makes it well-suited for spectra organized by wavenumber (Tan et al., 2022). In this study, LSTM models were implemented using an in-house Python script based on TensorFlow 1.12.0. Hyperparameter optimization was performed by varying the number of LSTM units (5, 10, 15, 20). The final architecture consisted of an LSTM layer, followed by a multilayer perceptron layer with 64 neurons and a ReLU activation function, and a sigmoid-activated output layer with 1 neuron for binary prediction. The models were trained using the Adam optimizer (learning rate = 0.0001) for 200 epochs with a batch size of 64.

Dataset. To investigate whether model performance improved as the diagnosis date approached, we split the data in the disease group, including preprocessed milk spectra, milk yield per milking, SCC, and DIM into 8 subsets based on days before the diagnosis date: >10 d, 10 to 8 d, 7 to 6 d, 5 to 4 d, 3 d, 2 d, 1 d and 0 d. Each of these subsets was combined with the full healthy group to construct separate predictive models. As described earlier, we used a double cross-validation framework with 50 rounds of random downsampling from the healthy group, matching the size and DIM distribution of each disease subset.

For each time point subset, we performed 3 sets of comparative analyses to evaluate model performance:

1. We compared the capabilities of preprocessed milk FTIR spectra, which describe the absorbance at each wave number, with their first-order derivatives (Savitzky–Golay first-order derivatives, 3 data points each side; Savitzky and Golay 1964;

De Marchi et al., 2009; De Marchi et al., 2011) for disease prediction;

2. We compared 2 spectral regions described previously: R1+R2+R3 and R1+R2+R3+R4, to explore whether R4 could convey useful information for disease prediction;
3. We examined whether milk FTIR spectra provide additional predictive value beyond readily available cow-level features that represent current industry standards by using PLS-DA models. To do so, we systematically compared models using (1) FTIR milk spectra alone, (2) commonly used cow-level features (milk yield, SCC, parity, and DIM) alone, (3) these cow-level features plus FTIR-derived milk composition traits (fat %, protein %, and lactose %; Callero et al., 2023), and (4) all features combined. The categorical feature “parity” was one-hot encoded into 3 binary variables (1, 2, or 3+) using Scikit-learn’s OneHotEncoder. Continuous features, including milk yield, SCC, DIM, fat %, protein %, and lactose %, were min-max normalized to a [0,1] range. All static features and spectral data, each scaled to the [0, 1] range, as previously described, were concatenated into a single input array to allow PLS-DA to capture the interaction among them.

Modeling Evaluation. To evaluate the statistical significance of model performance metrics—including AUROC, accuracy, sensitivity, and specificity—permutation tests (Lindgren et al., 1996; Westerhuis et al., 2008) were conducted. Under the null hypothesis (H_0), which assumes no association between features and labels (healthy or disease), sample labels were randomly permuted, and models were retrained on the permuted dataset. Evaluation metrics were calculated for each of the $n = 1,000$ permutations to generate a null distribution representing performance expected by chance. The observed metrics from the true (nonpermuted with correct group labels) data were then compared with the corresponding null distributions to derive P -values, calculated as:

$$P = \frac{1 + \#(metric_{perm} \gg metric_{true})}{N},$$

where $metric \in \{AUROC, Accuracy, Sensitivity, Specificity\}$ and $\#(metric_{perm} \gg metric_{true})$ is the number of permutations in which the permuted performance metric equals or exceeds the observed value. The performance metrics were defined as follows:

$$Accuracy = \frac{TP + TN}{TN + FP + TP + FN},$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}},$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}},$$

where true positive (TP) denotes the number of correctly identified milk samples from at-risk cows, false positive (FP) denotes the number of healthy milk samples that were misclassified as disease, true negative (TN) denotes the number of healthy milk samples that were correctly identified, and false negative (FN) denotes the number of milk samples from at-risk cows that were misclassified as healthy. Evaluation metrics were reported as the mean values obtained from repeated down-sampled double cross-validation, accompanied by 95% CI and *P*-values derived from permutation testing.

Identification of Important FTIR Spectral Regions

The importance score of each feature (each wavenumber across the FTIR spectrum), which represents how much each feature contributes to the disease-prediction task, was defined and tailored to each model type. Variable importance in projection (VIP) was used in the PLS-DA model (Oussama et al., 2012). For ridge regression models, the regression weights or coefficients represented the importance for each feature in the disease-prediction task (Heuer et al., 2001). Impurity importance (mean decrease impurity) was used to measure the importance of each feature for RF (de Santana et al., 2018). We applied “SHapley Additive exPlanations” (SHAP) proposed by Lundberg and Lee (2017) to interpret the feature importance in LSTM, and mean(|SHAP value|) to represent the average impact of each feature to the final prediction. All feature importance values were averaged over the repeated down-sampled double cross-validation framework to ensure unbiased estimates.

Statistical Analysis

Statistical significance was determined using nonparametric tests. Mann–Whitney U-tests (Mann and Whitney, 1947) were applied to compare cow-level features and absorbance values between healthy and disease groups. Multiple Mann–Whitney U-tests with Benjamini–Hochberg (BH) correction (Benjamini and Hochberg, 1995) were used to compare these features between the healthy group and each disease group at different time points. Wilcoxon rank-sum tests (Wilcoxon, 1945) with BH correction were used for comparisons of model performance, including comparisons between models incorporating different feature combinations, and different model types

trained on the same data set. Linear regressions were used for modeling the linear relationship (1) between the predictive milk components and their corresponding absorption peaks, and (2) absorbance differences and importance scores at each wavenumber. All statistical analyses were conducted using R v4.3.2. All codes for modeling, statistical analysis and visualization can be found here: <https://github.com/lindan1128/Milk-FTIR-Spectra-Health-Prediction-Project>.

RESULTS

Descriptive Statistics for Cows at Different Times Before Disease Diagnosis

In our final single-farm dataset spanning the first 30 DIM, parity distributions for the finally enrolled cows were 229 vs. 120 for parity 1, and 596 vs. 191 for parity ≥ 2 in the healthy and disease groups, respectively. Disease prevalence was 12.3% for ketosis, 10.1% for metritis, 1.3% for DA, and 10.3% for mastitis (Figure 1D). The disease group had lower milk yield per milking (21.6 ± 8.8 kg) than the healthy group (26.1 ± 8.5 kg; $P < 0.001$; Figure 1E), with yield declining over time and differing at nearly all prediagnosis time points (Figure 1E, and see Supplemental Table S1A [see Notes] for pairwise yield comparisons with *P*-values across time). Somatic cell count was also higher in the disease group (520 ± 872 vs. 230 ± 477 thousand cells/mL; $P < 0.001$; Figure 1F), and remained consistently elevated, may reflecting disease progression (Figure 1F, Supplemental Table S1B).

Increasing Fat and Decreasing Protein and Lactose Absorbance Peaks as Diagnosis Nears

Mean spectra across 4 subregions R1, R2, R3 and R4 (from 3,000 to 1,000 cm^{-1}) in both groups are depicted in Figure 2A. Absorbance peaks most indicative of fat, protein, and lactose components of milk were identified based on Heuer et al. (2001) and validated their positive associations with milk components as reported by Calero et al. (2023; Supplemental Figure S2, see Notes). The disease group showed higher absorbance peaks for fat (2,923 cm^{-1} , 2854 cm^{-1} , and 1,747 cm^{-1} : all $P < 0.001$) and lower absorbance peaks for protein (1,540 cm^{-1} : $P = 0.01$) and lactose (1,040 cm^{-1} : $P < 0.001$) than the healthy group. As the milk sampling date approached the disease diagnosis day, these differences became more pronounced, with fat peaks increasing and protein and lactose peaks decreasing in the disease group, all reaching their extrema on the diagnosis day (Figure 2B,

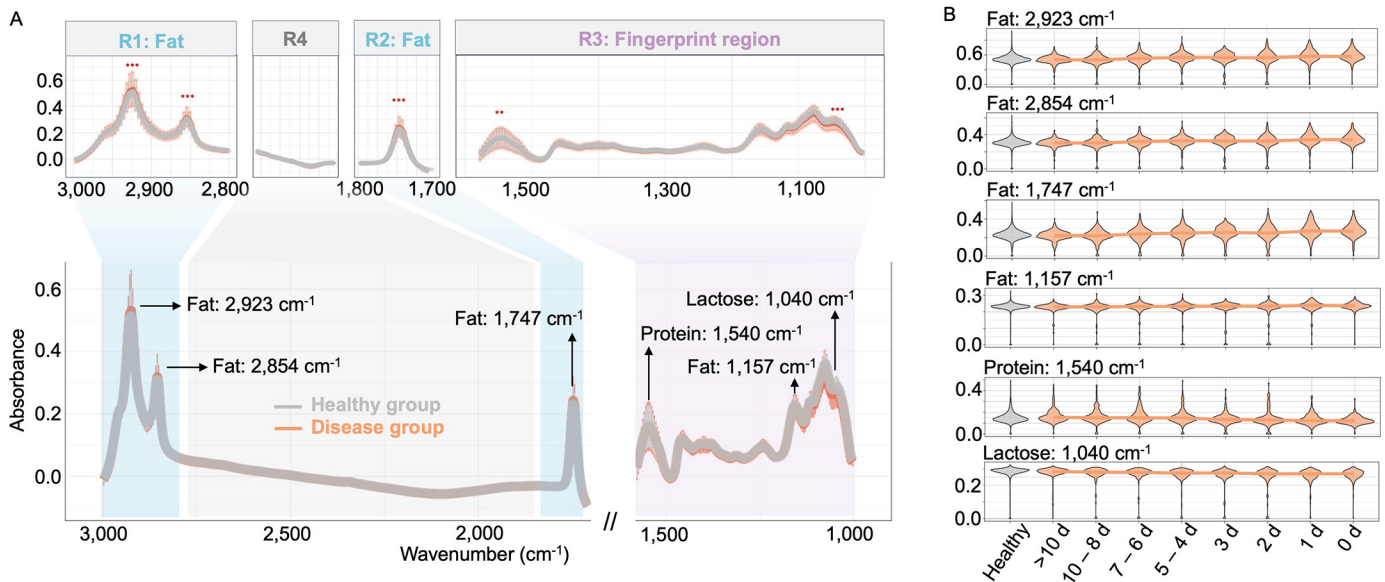


Figure 2. Milk spectra for healthy and disease groups. (A) Preprocessed milk FTIR spectra (mean $\pm$ SD; SD denoted by shaded area) and 4 subregions (R1, R2, R3, R4) for each group. (B) Absorbance of fat, protein, and lactose at specific wavenumbers in milk samples for the healthy and disease groups at various time points before disease diagnosis. *Adjusted $P < 0.05$. **Adjusted $P < 0.01$. ***Adjusted $P < 0.001$.

Supplemental Table S2 [see Notes] for detailed results including original and adjusted P -values).

Increasing Spectra-Based Predictive Performance Across Models as Diagnosis Nears

Spectra Across Different Times. Model performance improved consistently as the diagnosis date approached. With PLS-DA, AUROC rose from 0.55 (>10 d) to 0.78 (–1 d) and 0.83 (d 0) using R1+R2+R3, reflecting increases of 40.8% and 50.9%. Using R1+R2+R3+R4, AUROC rose from 0.59 to 0.78 and 0.82, representing 32.2% and 39.0% gains. Other performance metrics (accuracy, sensitivity, specificity) also improved as diagnosis approached in PLS-DA models using both R1+R2+R3 and R1+R2+R3+R4 spectral regions (Supplemental Figure S3, see Notes), with significance confirmed in almost all time periods, except for 10 to 8 d ($P = 0.18$ and 0.55, respectively), by permutation tests (1,000 iterations; all other $P < 0.05$; detailed metrics and test results in Supplemental Table S3, see Notes).

Spectra Versus Their First Derivatives. Using PLS-DA, models based on the first derivative of spectra achieved performance nearly identical to that of raw spectra, suggesting that both representations encode largely consistent information (Figure 3A). Models remained significant for most of time points (permutation test, 1,000 iterations; $P < 0.05$; Supplemental Table S3).

Spectra Versus Cow-Level Features. The PLS-DA models (based on different feature types) performance

improved as the diagnosis date approached (Figure 3B). Spectra-only models consistently outperformed models only using common cow-level features (DIM, parity, milk yield, SCC) from 7 d before diagnosis onward (Wilcoxon's rank-sum test with BH correction: all adjusted $P < 0.05$; Figure 3B and Supplemental Figure S4, see Notes), although they required more latent components to achieve optimal model performance (Supplemental Figure S5, see Notes). Adding FTIR spectroscopy-derived milk major components (fat %, protein %, lactose %) to common cow-level features improved their performance at several time points (d 5–4, 3, 1, and 0; all adjusted $P < 0.05$), but this combined model still underperformed compared with spectra-only models at multiple time points (7–4 d, 2 d, 0 d; all adjusted $P < 0.05$), suggesting these milk components only partially captured full spectral information. Combining all features (common cow-level features, spectra-derived milk major components and spectra) did not significantly outperform spectra-only models (all adjusted $P > 0.05$). However, this combined model did outperform models using common cow-level features plus spectra-derived components at 3 d, 2 d, and 0 d, and also outperformed common cow-level-only models from 7 d onward (all adjusted $P < 0.05$). These results indicated that although cow-level features capture some predictive signals, the full FTIR spectra provide additional information beyond what is available from these features. Supplemental Table S4 (see Notes) shows all detailed model performance with their permutation testing results.

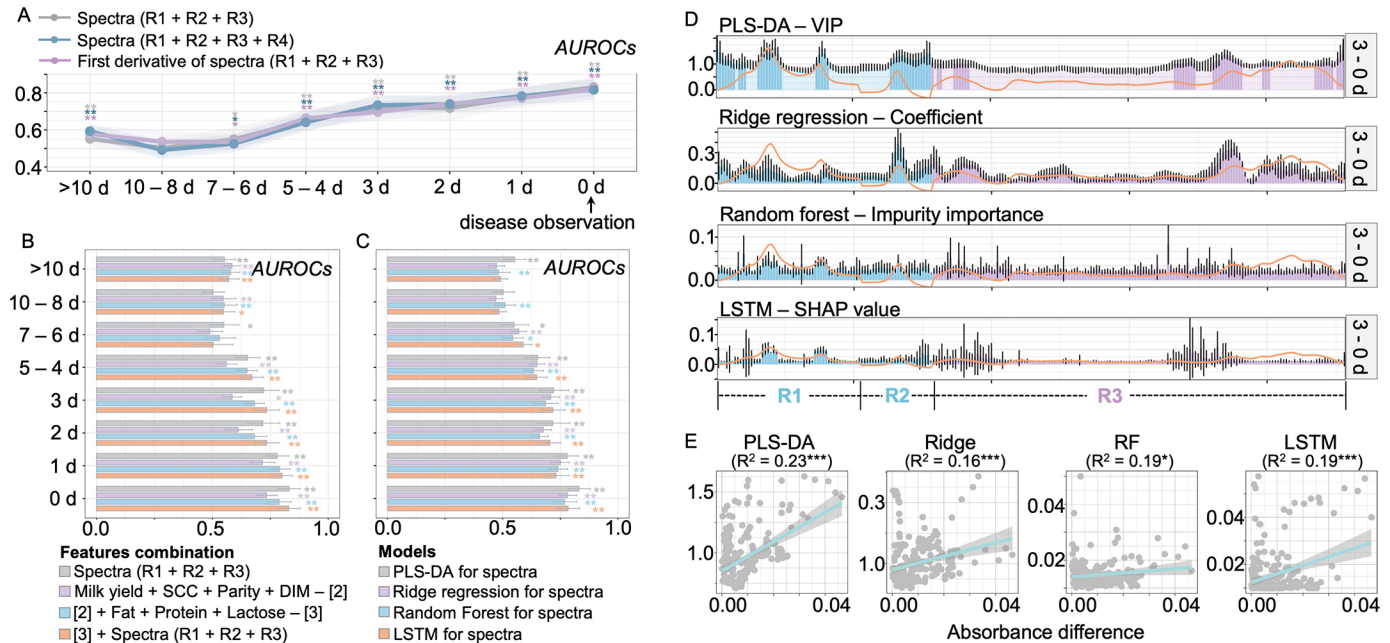


Figure 3. Comparison of predictive performance (AUROC score from outer validation; mean $\pm$ 95% CI) across models using different features at various time points before disease diagnosis. (A) Predictive performance of PLS-DA using milk FTIR spectra (R1+R2+R3 and R1+R2+R3+R4) and the first derivative. (B) Predictive performance of PLS-DA using milk FTIR spectra (R1+R2+R3) and various feature combinations (DIM, milk yield, SCC, parity, and FTIR spectra-derived milk fat %, protein %, and lactose %). (C) Predictive performance of various models (PLS-DA, ridge regression, RF, and LSTM) using milk FTIR spectra (R1+R2+R3). (D) Average feature importance for 4 models (PLS-DA, ridge regression, RF, and LSTM) over the 4 d before disease. For PLS-DA, only wavenumbers with average VIP >1 were highlighted. Mean spectra are overlaid in each plot. (E) Correlation analysis between the absolute value of difference in absorbance on each wavenumber between 2 groups and the importance score on each wavenumber for 4 models over the 4 d before disease detection. 1,000 times Permutation test: * $P < 0.05$. ** $P < 0.01$. *** $P < 0.001$.

Spectra with Different Models. Among all model types tested based on milk spectra, PLS-DA consistently outperformed others (Figure 3C and Supplemental Figure S6, see Notes). From 5 d before diagnosis, its performance was significantly higher than both ridge regression and RF (all adjusted $P < 0.05$; Figure 3C and Supplemental Figure S6). Although LSTM also surpassed ridge and RF from day -5 onward (all adjusted $P < 0.05$), it remained significantly inferior to PLS-DA on days -2 , -1 , and 0 (all adjusted $P < 0.05$). Supplemental Table S5 (see Notes) shows all detailed model results.

Correspondence Between Importance Scores and Group-Level Spectral Differences

Figure 3D shows the average importance of spectral regions in the disease prediction. All models consistently highlighted fat-associated absorption regions in R1 and R2 as key contributors, underscoring their diagnostic value. Similarly, R3, rich in protein, lactose and fat absorption signals, was consistently deemed informative, particularly within the ranges of $1,100$ to $1,000\text{ cm}^{-1}$ and $1,370$ to $1,260\text{ cm}^{-1}$.

We finally revealed the importance scores were closely aligned with spectral differences between groups across

all models (Figure 3E). Linear regression analysis confirmed associations between absorbance differences at each wavenumber and corresponding importance scores for all models. Specifically, PLS-DA, ridge regression, RF, and LSTM yielded R^2 of 0.23 , 0.16 , 0.19 , and 0.19 , respectively, with $P < 0.05$ for all.

DISCUSSION

Our study probed the potential of predicting the subsequent health status of dairy cows during early lactation by applying a series of classic machine and deep-learning techniques to milk FTIR spectra. Our results indicate that milk FTIR spectra might be a useful predictor for the early identification of sick cows. The prospect of using routinely collected milk samples represents an innovative, low-cost and, importantly, noninvasive tool. Such a tool would not only complement current clinical diagnostic methods, but also enable fast and informed management decisions that could improve the health and welfare of dairy cows in addition to reducing farm costs

Biological Relevance of Milk FT-Mid-Infrared Spectra for Early Disease Detection

Advancing beyond prior work based on FTIR-spectroscopic estimated milk components, our study showed that raw milk FTIR spectra capture unique diagnostic signals not fully represented by common cow-level features (e.g., DIM, milk yield, SCC, parity) or some FTIR spectroscopic-derived proxies (fat %, protein %, and lactose %). These findings indicated that mid-infrared (MIR) spectra encode complex biochemical information reflective of subclinical physiological alterations that precede overt disease. Our temporal tracking of representative absorbance peaks corresponding to fat, protein, and lactose (Figure 2B) revealed that deviations between healthy and diseased cows emerged multiple days before diagnosis, and their importance was also confirmed by different model types (Figure 3D–E).

This pattern aligned with established physiological mechanisms. When dairy cows become ill during the early-lactation period, they consume less feed, leading to inadequate energy intake and, consequently, reduced production of protein and lactose in milk (Reist et al., 2002; Negussie et al., 2013). At the same time, sick cows mobilize body fat reserves in the form of triglycerides to fulfill the energy requirement of lactation, resulting in a higher fat content in milk (de Vries and Veerkamp 2000; Reist et al., 2002). Thus, the alterations in milk components are directly reflected in the changes in milk spectra, thereby serving as indicators of the health status of dairy cows. Importantly, our current understanding on the relationships between milk MIR spectra and dominant milk components are still limited. As spectral regions associated with more milk constituents such as fatty acids and metabolic intermediates become better characterized, the diagnostic potential of milk MIR spectra will continue to expand. Perhaps in the future, a carefully selected subset of spectra-predicted components, combined with routinely collected cow-level information, may serve as a more interpretable and computationally efficient alternative to full-spectrum modeling, provided these features sufficiently capture the underlying diagnostic signals

Modeling Considerations in Milk FTIR-Based Disease Prediction

The predictive utility of milk FTIR spectra depends not only on the selection of wavenumber regions but also on preprocessing and modeling strategies (Rutten et al., 2009; Karoui et al., 2010; Ferragina et al., 2015). Although prior studies debated whether first-derivative spectra outperform raw absorbance values (De Marchi et al., 2009, 2011; Soyeurt et al., 2011), our results showed that both yielded nearly identical performance in PLS-

DA models. This suggested that major diagnostic signals are robust to such transformations, and that raw spectra already encapsulate the relevant biochemical variation without needing baseline correction

Although partial least squares regression and PLS-DA dominate quantitative and qualitative modeling due to their ability to handle multidimensional correlated spectral data (Soyeurt et al., 2020; Frizzarin et al., 2021), recent advances in machine learning offer greater flexibility and higher accuracy in modeling complex associations between milk spectra and targeted phenotype, surpassing PLS methods (Ferragina et al., 2015; Giannuzzi et al., 2023; Macedo Mota et al., 2023). In our study, within a sufficient parameter searching space (Supplemental Figure S5), PLS-DA consistently outperformed both regression- and tree-based models across most time points, highlighting its continued relevance. The strong performance of PLS-DA likely reflected its ability to extract latent variables most relevant to class discrimination, while maintaining interpretability and stability under modest sample sizes and high-dimensional input.

Deep learning, a special branch of machine learning, capitalizes on using large datasets to autonomously discover patterns, showing promise in several FTIR-based applications like pregnancy status prediction (Brand et al., 2021), tuberculosis status prediction (Denholm et al., 2020), milk lactoferrin content prediction (Soyeurt et al., 2020), and blood metabolic profile prediction (Pralle et al., 2018; Giannuzzi et al., 2023), by leveraging its capacity to learn complex, nonlinear patterns directly from raw data. Nevertheless, to the best of our knowledge, the application of deep-learning methods in predicting diseases using milk spectral data from FTIR spectroscopy is still very rare. In our study, the LSTM model, which received sequences of spectra from multiple days to mimic longitudinal variation, achieved competitive performance but did not always surpass PLS-DA. This may suggest that the discriminative signal within milk spectra is already well captured by latent variable models, and perhaps deep models require either larger datasets or greater input heterogeneity to fully exploit their potential. In addition, deep-learning methods, although powerful, are often more sensitive to shifts in data distribution and may not generalize as robustly to new herds or populations as classical approaches like PLS-DA, especially in limited-data settings. As the volume and diversity of milk spectral datasets expand, deep-learning frameworks may gain further advantage.

Practical Potentials and Implications

Although our analyses indicate that milk FTIR spectra may carry information relevant to health monitoring, translating any signal into practice will depend on op-

erational constraints and further evaluation, especially around sampling logistics, turnaround time, and actionable interventions. Under current workflow, FTIR analysis is conducted through central laboratories, leading to an approximately 1-to-3 day interval between sampling and results; this limits the utility of predictions for fast-progressing clinical diseases but may still support followup for more slowly developing problems. In this single-farm dataset, although we observed spectral deviations several days before clinical diagnosis, the consistency and lead time of such patterns should be established prospectively before informing routine decision-making.

We do not propose using FTIR-spectroscopic outputs to trigger immediate antibiotic treatment. Where validated, confidence-qualified predictions could support stepped responses (e.g., increased observation, repeat sampling, or targeted diagnostics), aligning with selective, data-informed health management and prudent antimicrobial use. Selective-prediction schemes that suppress low-confidence outputs may offer a way to balance sensitivity and actionability, but threshold settings and calibration would likely need to be farm-specific. Looking ahead, emerging portable or in-line MIR/near-IR systems may reduce turnaround time and cost. If such tools are adopted and validated, spectral monitoring could serve as one component within broader and multitiered decision-support workflows, rather than functioning as a stand-alone solution.

Limitations of this Study

This work has multiple limitations that merit attention. First, as mentioned previously, this analysis used data from a single farm with one management scheme, diet, season, and daytime sampling schedule. The limited sample size and homogeneous cohort constrain external validity, and transferability to other herds, regions, seasons, or operational practices was not evaluated. Second, practical deployment was also outside the scope of this work. The current pipeline was developed and assessed retrospectively and has not been tested for on-farm use cases (e.g., alert thresholds, lead time, false-positive burden, actionability, and costs). Third, “disease” is used here as a general concept; however, milk spectra from cows with distinct postpartum conditions (e.g., metritis and mastitis) may be heterogeneous, which can dilute model specificity and limit interpretability. Taken together, this study should be viewed as an initial, bounded evaluation intended to inform larger, disease-specific, multi-farm studies, and the potential for milk FTIR spectra to predict upcoming postpartum disease in early-lactation Holstein dairy cows observed in this single-farm context should be considered hypothesis-generating until externally validated and prospectively shown to have clinical utility.

CONCLUSIONS

We retrospectively evaluated the potential of disease prediction in early-lactation Holstein dairy cows using milk FTIR spectra and different machine learning and deep-learning models. Our results demonstrated that early-lactation milk FTIR spectra capture physiologically meaningful changes in fat, protein, and lactose absorbance that precede clinical diagnosis. These spectral patterns were detected by regression, machine learning, and deep-learning approaches and showed encouraging within-dataset performance. This work should be regarded as an initial step in assessing whether milk FTIR can inform disease risk assessment, rather than as evidence for immediate large-scale deployment or economic impact. We aim to increase the sample size in the future to increase the robustness of the conclusions and improve the performance and utility of the disease-prediction models.

NOTES

This work was supported by the USDA (Washington, DC) National Institute of Food and Agriculture Multi-state project no. 1023396 and Hatch project no. 7000969 awarded to J. A. A. M. and the Health and Medical Research Fund (10213386), Guangdong Basic and Applied Research Major Program (2019B030302005), and City University of Hong Kong internal grant (7005880 and 9680310) awarded to J. L. The authors would like to thank the participating farm, Dave Barbano and Chasidy Coon (Cornell University, Ithaca, NY) for their assistance in milk analysis, and Katie Callero, Ian Frost, Regina Martinez, Hannah McCray, and Abigail Reed (Cornell University, Ithaca, NY) for their help with collection of milk samples. Supplemental material for this article is available at <https://zenodo.org/records/15776317>. Because our study did not involve direct use of animals, no animal care and use approval was required; ethical and scientific review and was approved by the Cornell University Veterinary Clinical Studies Committee (CUVCSC Protocol ID#062420-05). The authors declare the following relevant conflict of interest: one of the co-corresponding authors, J. A. A. M., serves as the Editor in Chief of *JDS Communications*. The authors have not stated any other conflicts of interest.

Nonstandard abbreviations used: AUROC = area under the receiver operating characteristic curve, BH = Benjamini–Hochberg; DAb = displaced abomasum, FTIR = fourier-transform infrared, In. = inner; LSTM = long short-term memory, MIR = mid-infrared; Out. = outer; PLS-DA = partial least squares-discriminant analysis, RF = random forest, RNN = recurrent neural network; Train

= training; Val = validation; VIP = variable importance in projection.

REFERENCES

- Bach, K. D., D. M. Barbano, and J. A. A. McArt. 2019. Association of mid-infrared-predicted milk and blood constituents with early-lactation disease, removal, and production outcomes in Holstein cows. *J. Dairy Sci.* 102:10129–10139. <https://doi.org/10.3168/jds.2019-16926>.
- Bach, K. D., D. M. Barbano, and J. A. A. McArt. 2021. The relationship of excessive energy deficit with milk somatic cell score and clinical mastitis. *J. Dairy Sci.* 104:715–727. <https://doi.org/10.3168/jds.2020-18432>.
- Barker, M., and W. Rayens. 2003. Partial least squares for discrimination. *J. Chemometr.* 17:166–173. <https://doi.org/10.1002/cem.785>.
- Benjamini, Y., and Y. Hochberg. 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *J. R. Stat. Soc. Series B Stat. Methodol.* 57:289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.
- Bonfatti, V., S.-A. Turner, B. Kuhn-Sherlock, T. D. W. Luke, P. N. Ho, C. V. C. Phyn, and J. E. Pryce. 2019. Prediction of blood β -hydroxybutyrate content and occurrence of hyperketonemia in early-lactation, pasture-grazed dairy cows using milk infrared spectra. *J. Dairy Sci.* 102:6466–6476. <https://doi.org/10.3168/jds.2018-15988>.
- Brand, W., A. T. Wells, S. L. Smith, S. J. Denholm, E. Wall, and M. P. Coffey. 2021. Predicting pregnancy status from mid-infrared spectroscopy in dairy cow milk using deep learning. *J. Dairy Sci.* 104:4980–4990. <https://doi.org/10.3168/jds.2020-18367>.
- Breiman, L. 2001. Random forests. *Mach. Learn.* 45:5–32. <https://doi.org/10.1023/A:1010933404324>.
- Callero, K. R., E. M. Teplitz, D. M. Barbano, C. R. Seely, J. A. Seminara, I. R. Frost, H. A. McCray, R. M. Martinez, A. M. Reid, and J. A. A. McArt. 2023. Patterns of Fourier-transform infrared estimated milk constituents in early lactation Holstein cows on a single New York State dairy. *J. Dairy Sci.* 106:2716–2728. <https://doi.org/10.3168/jds.2022-22588>.
- Chapinal, N., M. Carson, T. F. Duffield, M. Capel, S. Godden, M. Overton, J. E. P. Santos, and S. J. LeBlanc. 2011. The association of serum metabolites with clinical disease during the transition period. *J. Dairy Sci.* 94:4897–4903. <https://doi.org/10.3168/jds.2010-4075>.
- Cho, K., B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio. 2014. Learning phrase representations using RNN encoder-decoder for statistical machine translation. Pages 1724–1734 in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Doha, Qatar. Association for Computational Linguistics. <https://doi.org/10.3115/v1/D14-1179>.
- Churakov, M., J. Karlsson, A. Edvardsson Rasmussen, and K. Holtenius. 2021. Milk fatty acids as indicators of negative energy balance of dairy cows in early lactation. *Animal* 15:100253. <https://doi.org/10.1016/j.animal.2021.100253>.
- Dale, L. M., A. Werner, and F. Gollé-Leidreiter. 2019. ‘MastiMIR’-A mastitis early warning system based on MIR spectra. ICAR Technical Series. Rome, Italy.
- De Marchi, M., C. C. Fagan, C. P. O’Donnell, A. Cecchinato, R. Dal Zotto, M. Cassandro, M. Penasa, and G. Bittante. 2009. Prediction of coagulation properties, titratable acidity, and pH of bovine milk using mid-infrared spectroscopy. *J. Dairy Sci.* 92:423–432. <https://doi.org/10.3168/jds.2008-1163>.
- De Marchi, M., M. Penasa, A. Cecchinato, M. Mele, P. Secchiari, and G. Bittante. 2011. Effectiveness of mid-infrared spectroscopy to predict fatty acid composition of Brown Swiss bovine milk. *Animal* 5:1653–1658. <https://doi.org/10.1017/S1751731111000747>.
- De Marchi, M., V. Toffanin, M. Cassandro, and M. Penasa. 2014. *Invited review*: Mid-infrared spectroscopy as phenotyping tool for milk traits. *J. Dairy Sci.* 97:1171–1186. <https://doi.org/10.3168/jds.2013-6799>.
- Denholm, S. J., W. Brand, A. P. Mitchell, A. T. Wells, T. Krzyzelewski, S. L. Smith, E. Wall, and M. P. Coffey. 2020. Predicting bovine tuberculosis status of dairy cows from mid-infrared spectral data of milk using deep learning. *J. Dairy Sci.* 103:9355–9367. <https://doi.org/10.3168/jds.2020-18328>.
- de Santana, F. B., S. J. Mazivila, L. C. Gontijo, W. B. Neto, and R. J. Poppi. 2018. Rapid discrimination between authentic and adulterated andiroba oil using FTIR-HATR spectroscopy and random forest. *Food Anal. Methods* 11:1927–1935. <https://doi.org/10.1007/s12161-017-1142-5>.
- de Vries, M. J., and R. F. Veerkamp. 2000. Energy balance of dairy cattle in relation to milk production variables and fertility. *J. Dairy Sci.* 83:62–69. [https://doi.org/10.3168/jds.S0022-0302\(00\)74856-9](https://doi.org/10.3168/jds.S0022-0302(00)74856-9).
- Ferragina, A., G. De Los Campos, A. I. Vazquez, A. Cecchinato, and G. Bittante. 2015. Bayesian regression models outperform partial least squares methods for predicting milk components and technological properties using infrared spectral data. *J. Dairy Sci.* 98:8133–8151. <https://doi.org/10.3168/jds.2014-9143>.
- Filzmoser, P., B. Liebmann, and K. Varmuza. 2009. Repeated double cross validation. *J. Chemometr.* 23:160–171. <https://doi.org/10.1002/cem.1225>.
- Frizzarin, M., I. C. Gormley, D. P. Berry, T. B. Murphy, A. Casa, A. Lynch, and S. McParland. 2021. Predicting cow milk quality traits from routinely available milk spectra using statistical machine learning methods. *J. Dairy Sci.* 104:7438–7447. <https://doi.org/10.3168/jds.2020-19576>.
- Giannuzzi, D., L. F. M. Mota, S. Pegolo, F. Tagliapietra, S. Schiavon, L. Gallo, P. A. Marsan, E. Trevisi, and A. Cecchinato. 2023. Prediction of detailed blood metabolic profile using milk infrared spectra and machine learning methods in dairy cattle. *J. Dairy Sci.* 106:3321–3344. <https://doi.org/10.3168/jds.2022-22454>.
- Grelet, C., J. A. Fernández Pierna, P. Dardenne, V. Baeten, and F. Dehareng. 2015. Standardization of milk mid-infrared spectra from a European dairy network. *J. Dairy Sci.* 98:2150–2160. <https://doi.org/10.3168/jds.2014-8764>.
- Gröhn, Y. T. 2000. Milk yield and disease: Towards optimizing dairy herd health and management decisions. *Bov. Pract.* 34:32–40. <https://doi.org/10.21423/bovine-vol34no1p32-40>.
- Gross, J. J., and R. M. Bruckmaier. 2019. Review: Metabolic challenges in lactating dairy cows and their assessment via established and novel indicators in milk. *Animal* 13:s75–s81. <https://doi.org/10.1017/S175173111800349X>.
- Hamann, J., and V. Krömker. 1997. Potential of specific milk composition variables for cow health management. *Livest. Prod. Sci.* 48:201–208. [https://doi.org/10.1016/S0301-6226\(97\)00027-4](https://doi.org/10.1016/S0301-6226(97)00027-4).
- Hammon, D. S., I. M. Evjen, T. R. Dhiman, J. P. Goff, and J. L. Walters. 2006. Neutrophil function and energy status in Holstein cows with uterine health disorders. *Vet. Immunol. Immunopathol.* 113:21–29. <https://doi.org/10.1016/j.vetimm.2006.03.022>.
- Hertl, J. A., Y. H. Schukken, L. W. Tauer, F. L. Welcome, and Y. T. Gröhn. 2018. Does clinical mastitis in the first 100 days of lactation 1 predict increased mastitis occurrence and shorter herd life in dairy cows? *J. Dairy Sci.* 101:2309–2323. <https://doi.org/10.3168/jds.2017-12615>.
- Heuer, C., H. J. Luinge, E. T. G. Lutz, Y. H. Schukken, J. H. Van Der Maas, H. Wilmink, and J. P. T. M. Noordhuizen. 2001. Determination of acetone in cow milk by Fourier transform infrared spectroscopy for the detection of subclinical ketosis. *J. Dairy Sci.* 84:575–582. [https://doi.org/10.3168/jds.S0022-0302\(01\)74510-9](https://doi.org/10.3168/jds.S0022-0302(01)74510-9).
- Hochreiter, S., and J. Schmidhuber. 1997. Long short-term memory. *Neural Comput.* 9:1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>.
- Hoerl, A. E., and R. W. Kennard. 1970. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics* 12:55–67. <https://doi.org/10.1080/00401706.1970.10488634>.
- Karoui, R., G. Downey, and C. Blecker. 2010. Mid-Infrared spectroscopy coupled with chemometrics: A tool for the analysis of intact food systems and the exploration of their molecular structure–quality relationships—A review. *Chem. Rev.* 110:6144–6168. <https://doi.org/10.1021/cr100090k>.

- Leblanc, S. 2010. Monitoring metabolic health of dairy cattle in the transition period. *J. Reprod. Dev.* 56:S29–S35. <https://doi.org/10.1262/jrd.1056S29>.
- Lindgren, F., B. Hansen, W. Karcher, M. Sjöström, and L. Eriksson. 1996. Model validation by permutation tests: Applications to variable selection. *J. Chemom.* 10:521–532. [https://doi.org/10.1002/\(SICI\)1099-128X\(199609\)10:5<521::AID-CEM448>3.0.CO;2-J](https://doi.org/10.1002/(SICI)1099-128X(199609)10:5<521::AID-CEM448>3.0.CO;2-J).
- Luinge, H. J., E. Hop, E. T. G. Lutz, J. A. Van Hemert, and E. A. M. De Jong. 1993. Determination of the fat, protein and lactose content of milk using Fourier transform infrared spectrometry. *Anal. Chim. Acta* 284:419–433. [https://doi.org/10.1016/0003-2670\(93\)85328-H](https://doi.org/10.1016/0003-2670(93)85328-H).
- Lundberg, S. M., and S. I. Lee. 2017. A unified approach to interpreting model predictions. Pages XXX–XXX in Proceedings of the 31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA. <https://proceedings.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf>.
- Macedo Mota, L. F., V. Bisutti, A. Vanzin, S. Pegolo, A. Toscano, S. Schiavon, F. Tagliapietra, L. Gallo, P. Ajmone Marsan, and A. Cecchinato. 2023. Predicting milk protein fractions using infrared spectroscopy and a gradient boosting machine for breeding purposes in Holstein cattle. *J. Dairy Sci.* 106:1853–1873. <https://doi.org/10.3168/jds.2022-22119>.
- Mann, H. B., and D. R. Whitney. 1947. On a test of whether one of two random variables is stochastically larger than the other. *Ann. Math. Stat.* 18:50–60. <https://doi.org/10.1214/aoms/1177730491>.
- McArt, J. A. A., D. V. Nydam, and G. R. Oetzel. 2012. Epidemiology of subclinical ketosis in early lactation dairy cattle. *J. Dairy Sci.* 95:5056–5066. <https://doi.org/10.3168/jds.2012-5443>.
- McParland, S., G. Banos, E. Wall, M. P. Coffey, H. Soyeurt, R. F. Veerkamp, and D. P. Berry. 2011. The use of mid-infrared spectrometry to predict body energy status of Holstein cows. *J. Dairy Sci.* 94:3651–3661. <https://doi.org/10.3168/jds.2010-3965>.
- McParland, S., E. Lewis, E. Kennedy, S. G. Moore, B. McCarthy, M. O'Donovan, S. T. Butler, J. E. Pryce, and D. P. Berry. 2014. Mid-infrared spectrometry of milk as a predictor of energy intake and efficiency in lactating dairy cows. *J. Dairy Sci.* 97:5863–5871. <https://doi.org/10.3168/jds.2014-8214>.
- Negussie, E., I. Strandén, and E. A. Mäntysaari. 2013. Genetic associations of test-day fat:protein ratio with milk yield, fertility, and udder health traits in Nordic Red cattle. *J. Dairy Sci.* 96:1237–1250. <https://doi.org/10.3168/jds.2012-5720>.
- Oussama, A., F. Elabadi, S. Platikanov, F. Kzaiber, and R. Tauler. 2012. Detection of olive oil adulteration using FT-IR spectroscopy and PLS with variable importance of projection (VIP) scores. *J. Am. Oil Chem. Soc.* 89:1807–1812. <https://doi.org/10.1007/s11746-012-2091-1>.
- Overton, T. R., J. A. A. McArt, and D. V. Nydam. 2017. *A 100-Year Review: Metabolic health indicators and management of dairy cattle*. *J. Dairy Sci.* 100:10398–10417. <https://doi.org/10.3168/jds.2017-13054>.
- Pazzola, M., G. Stocco, A. Ferragina, G. Bittante, M. L. Dettori, G. M. Vacca, and C. Cipolat-Gotet. 2023. Cheese yield and nutrients recovery in the curd predicted by Fourier-transform spectra from individual sheep milk samples. *J. Dairy Sci.* 106:6759–6770. <https://doi.org/10.3168/jds.2023-23349>.
- Pedregosa, F., G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, A. Müller, J. Nothman, G. Louppe, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay. 2012. Scikit-learn: Machine learning in Python. <https://doi.org/10.48550/ARXIV.1201.0490>.
- Pralle, R. S., K. W. Weigel, and H. M. White. 2018. Predicting blood β -hydroxybutyrate using milk Fourier transform infrared spectrum, milk composition, and producer-reported variables with multiple linear regression, partial least squares regression, and artificial neural network. *J. Dairy Sci.* 101:4378–4387. <https://doi.org/10.3168/jds.2017-14076>.
- Reist, M., D. Erdin, D. Von Euw, K. Tschuempferlin, H. Leuenberger, Y. Chilliard, H. M. Hammon, C. Morel, C. Philipona, Y. Zbinden, N. Kuenzi, and J. W. Blum. 2002. Estimation of energy balance at the individual and herd level using blood and milk traits in high-yielding dairy cows. *J. Dairy Sci.* 85:3314–3327. [https://doi.org/10.3168/jds.S0022-0302\(02\)74420-2](https://doi.org/10.3168/jds.S0022-0302(02)74420-2).
- Rieness, L., N. Khayatdadeh, A. Köck, C. Egger-Danner, N. Gengler, C. Grelet, L. M. Dale, A. Werner, F.-J. Auer, J. Leblois, and J. Sölkner. 2022. Prediction of acute and chronic mastitis in dairy cows based on somatic cell score and mid-infrared spectroscopy of milk. *Animals (Basel)* 12:1830. <https://doi.org/10.3390/ani12141830>.
- Rutten, M. J. M., H. Bovenhuis, K. A. Hettinga, H. J. F. Van Valenberg, and J. A. M. Van Arendonk. 2009. Predicting bovine milk fat composition using infrared spectroscopy based on milk samples collected in winter and summer. *J. Dairy Sci.* 92:6202–6209. <https://doi.org/10.3168/jds.2009-2456>.
- Sargeant, J. M., A. M. O'Connor, I. R. Dohoo, H. N. Erb, M. Cevallos, M. Egger, A. K. Ersbøll, S. W. Martin, L. R. Nielsen, D. L. Pearl, D. U. Pfeiffer, J. Sanchez, M. E. Torrence, H. Vigre, C. Waldner, and M. P. Ward. 2016. Methods and processes of developing the strengthening the reporting of observational studies in epidemiology—Veterinary (STROBE-Vet) statement. *Prev. Vet. Med.* 134:188–196. <https://doi.org/10.1016/j.prevetmed.2016.09.005>.
- Savitzky, A., and M. J. E. Golay. 1964. Smoothing and differentiation of data by simplified least squares procedures. *Anal. Chem.* 36:1627–1639. <https://doi.org/10.1021/ac60214a047>.
- Seminara, J. A., K. R. Callero, I. R. Frost, R. M. Martinez, H. A. McCray, A. M. Reid, C. R. Seely, D. M. Barbano, and J. A. A. McArt. 2023. Calcium dynamics and associated temporal patterns of milk constituents in early-lactation multiparous Holsteins. *J. Dairy Sci.* 106:7117–7130. <https://doi.org/10.3168/jds.2022-23142>.
- Soyeurt, H., C. Bastin, F. G. Colinet, V. M.-R. Arnould, D. P. Berry, E. Wall, F. Dehareng, H. N. Nguyen, P. Dardenne, J. Schefers, J. Vandenplas, K. Weigel, M. Coffey, L. Théron, J. Detilleux, E. Reding, N. Gengler, and S. McParland. 2012. Mid-infrared prediction of lactoferrin content in bovine milk: Potential indicator of mastitis. *Animal* 6:1830–1838. <https://doi.org/10.1017/S1751731112000791>.
- Soyeurt, H., F. Dehareng, N. Gengler, S. McParland, E. Wall, D. P. Berry, M. Coffey, and P. Dardenne. 2011. Mid-infrared prediction of bovine milk fatty acids across multiple breeds, production systems, and countries. *J. Dairy Sci.* 94:1657–1667. <https://doi.org/10.3168/jds.2010-3408>.
- Soyeurt, H., C. Grelet, S. McParland, M. Calmels, M. Coffey, A. Tedde, P. Delhez, F. Dehareng, and N. Gengler. 2020. A comparison of 4 different machine learning algorithms to predict lactoferrin content in bovine milk from mid-infrared spectra. *J. Dairy Sci.* 103:11585–11596. <https://doi.org/10.3168/jds.2020-18870>.
- Subramanian, A., W. J. Harper, and L. E. Rodriguez-Saona. 2009. Cheddar cheese classification based on flavor quality using a novel extraction method and Fourier transform infrared spectroscopy. *J. Dairy Sci.* 92:87–94. <https://doi.org/10.3168/jds.2008-1449>.
- Susan, S., and A. Kumar. 2021. The balancing trick: Optimized sampling of imbalanced datasets—A brief survey of the recent state of the art. *Eng. Rep.* 3:e12298. <https://doi.org/10.1002/eng2.12298>.
- Szymańska, E., E. Saccenti, A. K. Smilde, and J. A. Westerhuis. 2012. Double-check: Validation of diagnostic statistics for PLS-DA models in metabolomics studies. *Metabolomics* 8(S1):3–16. <https://doi.org/10.1007/s11306-011-0330-3>.
- Tan, A., Y. Wang, Y. Zhao, B. Wang, X. Li, and A. X. Wang. 2022. Near infrared spectroscopy quantification based on Bi-LSTM and transfer learning for new scenarios. *Spectrochim. Acta A Mol. Biomol. Spectrosc.* 283:121759. <https://doi.org/10.1016/j.saa.2022.121759>.
- Tyagi, S., and S. Mittal. 2020. Sampling approaches for imbalanced data classification problem in machine learning. Pages 209–221 in Proceedings of ICRIC 2019. P. K. Singh, A. K. Kar, Y. Singh, M. H. Kolekar, and S. Tanwar, ed. Springer International Publishing, Cham. https://doi.org/10.1007/978-3-030-29407-6_17.
- van der Drift, S. G. A., K. J. E. Van Hulzen, T. G. Teweldemedhn, R. Jorritsma, M. Nielen, and H. C. M. Heuven. 2012. Genetic and non-genetic variation in plasma and milk β -hydroxybutyrate and milk acetone concentrations of early-lactation dairy cows. *J. Dairy Sci.* 95:6781–6787. <https://doi.org/10.3168/jds.2012-5640>.

- van Kneegsel, A. T. M., S. G. A. van der Drift, M. Horneman, A. P. W. De Roos, B. Kemp, and E. A. M. Graat. 2010. *Short communication: Ketone body concentration in milk determined by Fourier transform infrared spectroscopy: Value for the detection of hyperketonemia in dairy cows*. *J. Dairy Sci.* 93:3065–3069. <https://doi.org/10.3168/jds.2009-2847>.
- Westerhuis, J. A., H. C. Hoefsloot, S. Smit, D. J. Vis, A. K. Smilde, E. J. van Velzen, J. P. M. van Duijnhoven, and F. A. van Dorsten. 2008. Assessment of PLSDA cross validation. *Metabolomics* 4:81–89. <https://doi.org/10.1007/s11306-007-0099-6>.
- Wilcoxon, F. 1945. Individual comparisons by ranking methods. *Biometrics Bull.* 1:80–83. <https://doi.org/10.2307/3001968>.

ORCIDS

- D. Lin,  <https://orcid.org/0000-0001-8064-5666>
 J. A. Seminara,  <https://orcid.org/0000-0002-2933-7331>
 J. A. A. McArt  <https://orcid.org/0000-0001-5654-9172>